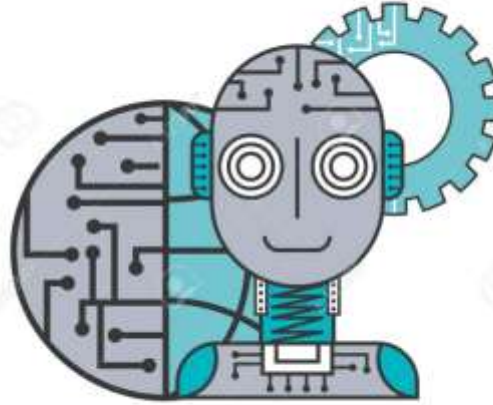




Introduction Frontiers in AI and Data Science

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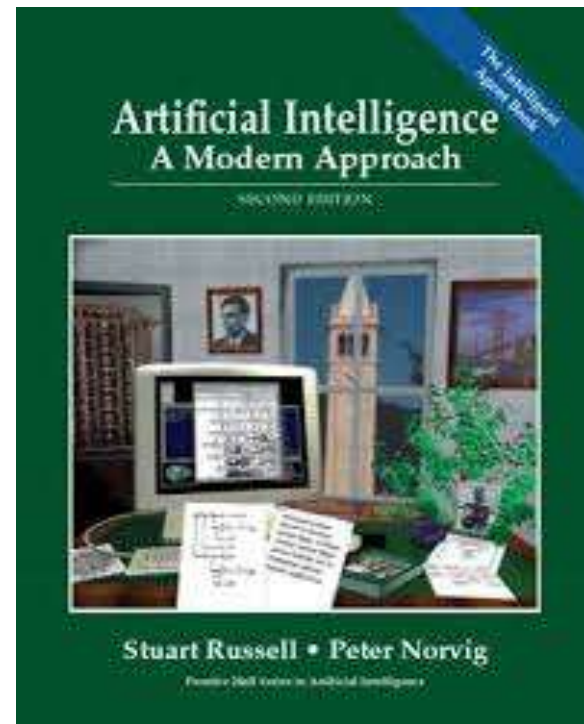
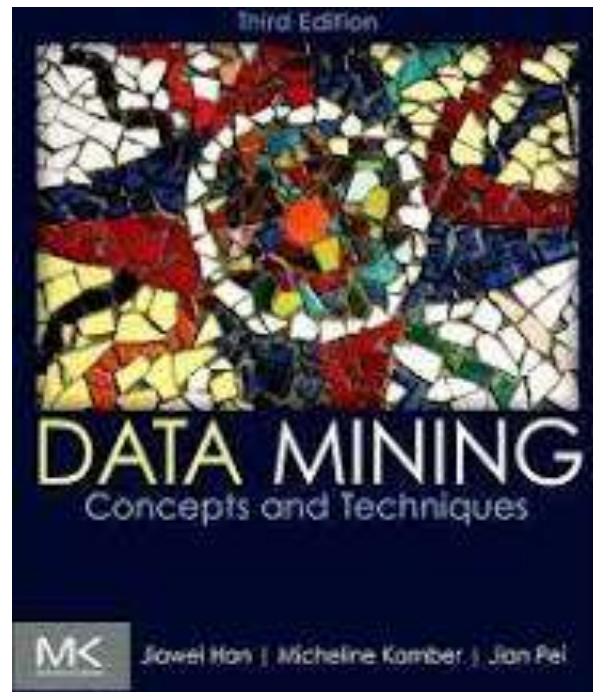
Outlook 1

- Core textbooks
- Module introduction



Sensitivity:
Internal

Core textbooks



Online Reading List ▼

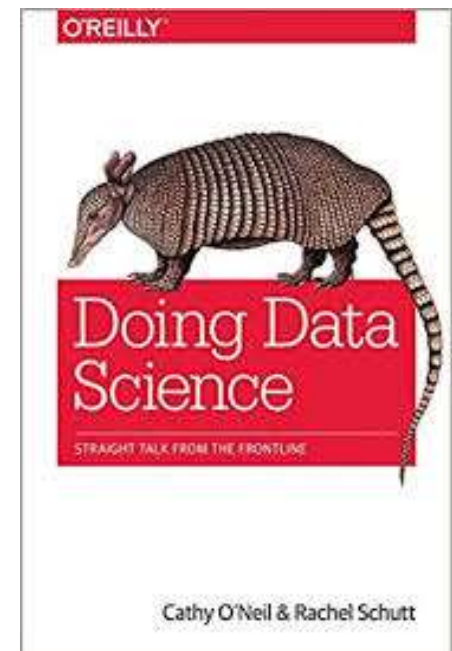
Build Content ▼

Assessments ▼

Add Interactive T



Three core textbooks of this module ▼



Module topics (Teaching Weeks)

Weeks	Theory	Practical
1	AI history, branches, application scenarios and relations to Data Mining and Machine Learning	Python basics
2	Data Pre-processing and EDA	Python - Data Pre-processing
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➡ A1 Deadline

➡ A2 Deadline

Learning outcomes

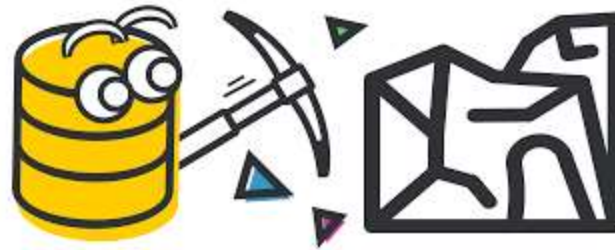
MLO1: Explain fundamental concepts in ML/ AI

MLO2: Apply data mining tools to a problem instance and interpret the results

MLO3: Select and justify choice of AI or ML approach for a specific problem instance

Outlook 2

- The definition of Data Mining
- The definition of AI and relations to Data Mining
- A bit of history of AI and branches
- Different level of Machine Intelligent (a demo)
- Turing test demo for fun
- Self-Reading Homework



Data Mining

What is Data Mining?

Data mining is a process to **turn raw data into useful information and knowledge** by using software and algorithms to look for **patterns** in **large batches of data**.

For example: businesses can learn more about their customers to develop more effective marketing strategies, increase sales and decrease costs.



Sensitivity:
Internal



Bread



Milk

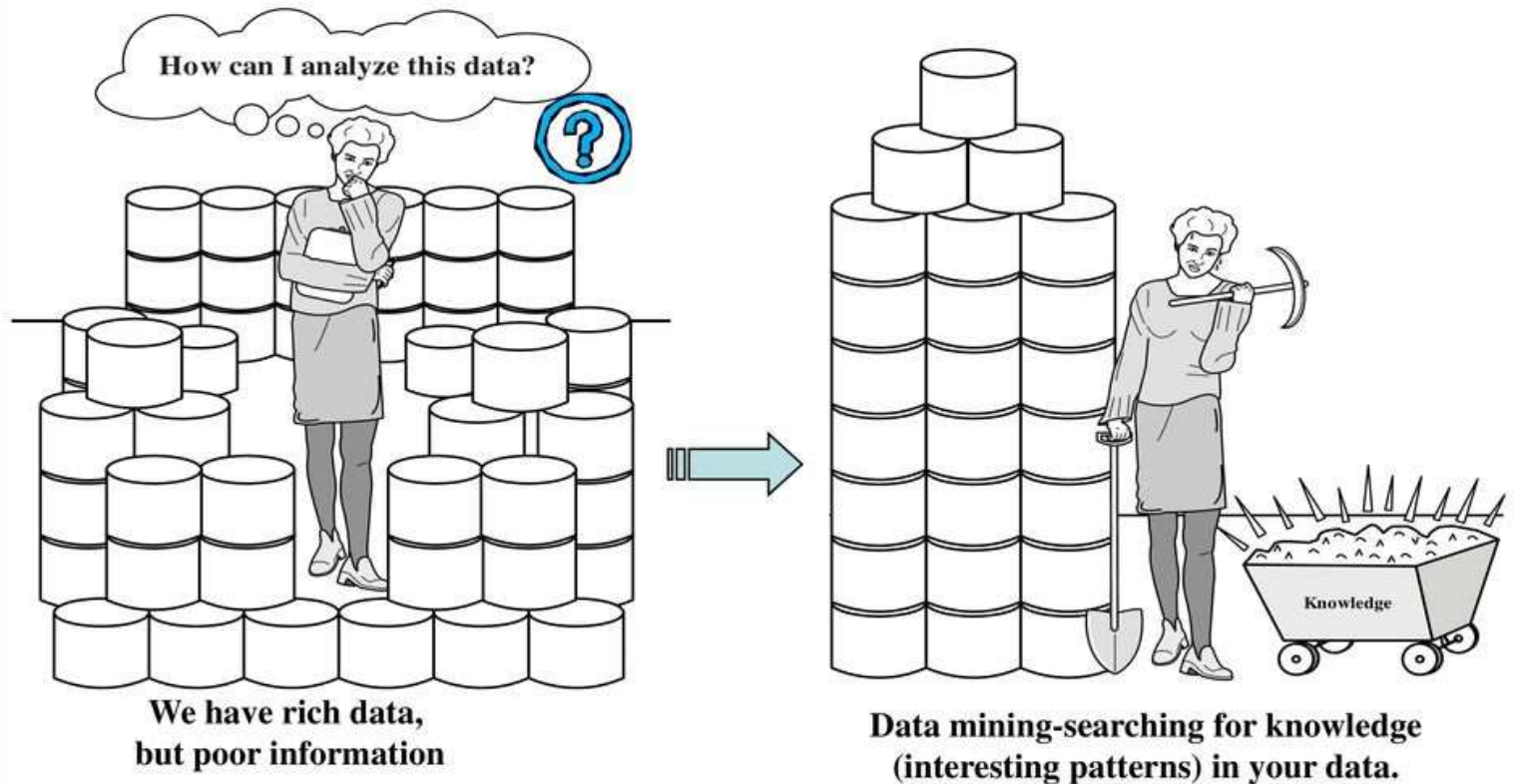


Butter

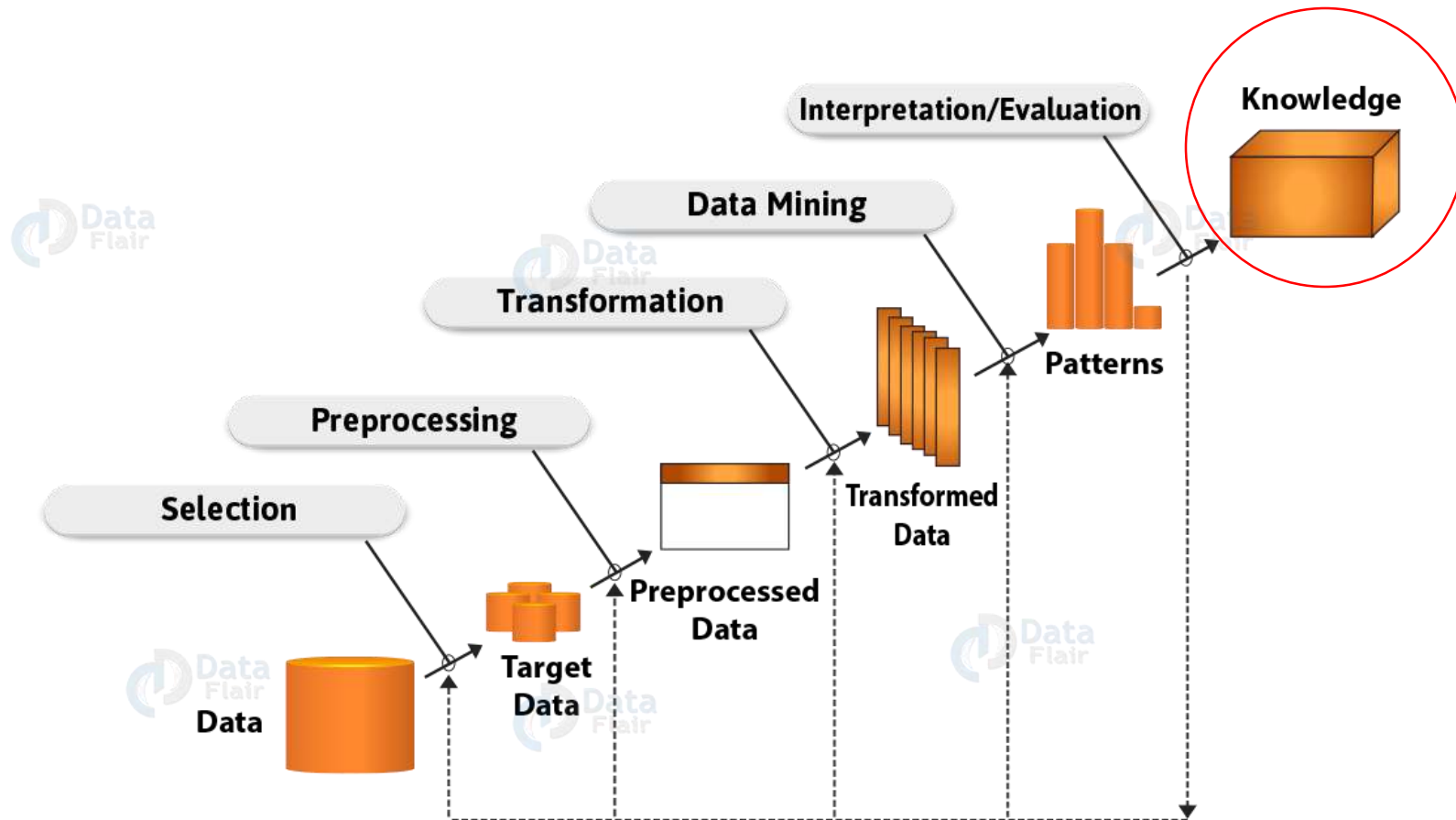


Aim of Data Mining

Data Mining



An important step of knowledge discovery



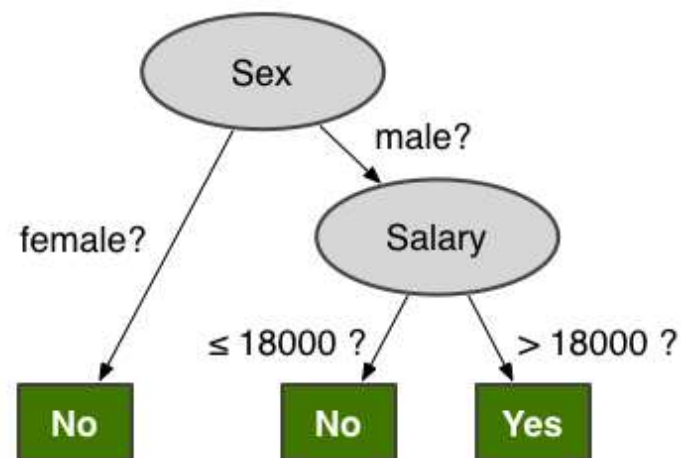
Data Mining example

Classification Target

Attributes

Instances

Name	Salary	Sex	Age	Buy widget
Bloggs	15000	male	19	No
Jones	25000	male	33	Yes
Smit	23000	female	50	No
Smit	16000	male	40	No
Smit	200	male	10	No
Patel	30000	female	30	No
Steel	25000	male	23	Yes
Higgs	18000	female	55	No
Puggs	50000	male	57	Yes
Puggs	51000	female	57	No





Artificial Intelligent (AI)

What is AI?

Artificial Intelligence, A Morden Approach. Second Edition

"The exciting new effort to make computers think . . . <i>machines with minds</i>, in the full and literal sense" (Haugeland, 1985) "[The automation of] activities that we associate with human thinking, activities such as decision-making, problem solving, learning . . ." (Bellman, 1978)	"The study of mental faculties through the use of computational models" (Charniak and McDermott, 1985) "The study of the computations that make it possible to perceive, reason, and act" (Winston, 1992)
"The art of creating machines that perform functions that require intelligence when performed by people" (Kurzweil, 1990) "The study of how to make computers do things at which, at the moment, people are better" (Rich and Knight, 1991)	"A field of study that seeks to explain and emulate intelligent behavior in terms of computational processes" (Schalkoff, 1990) "The branch of computer science that is concerned with the automation of intelligent behavior" (Luger and Stubblefield, 1993)

Systems that think like humans

Systems that think rationally

Systems that act like humans

Systems that act rationally

Morden or Your Definition

“AI refers to the simulation of human intelligence in machines that are programmed to think like humans and mimic their actions. The term may also be applied to any machine that exhibits traits associated with a human mind such as learning and problem-solving.”

- Investopedia

AI refers to a soft or hard device that can make inferences and best decisions to support next action or provide recommendations through collected digital information and humankind learning process.

- Me

The title of the module

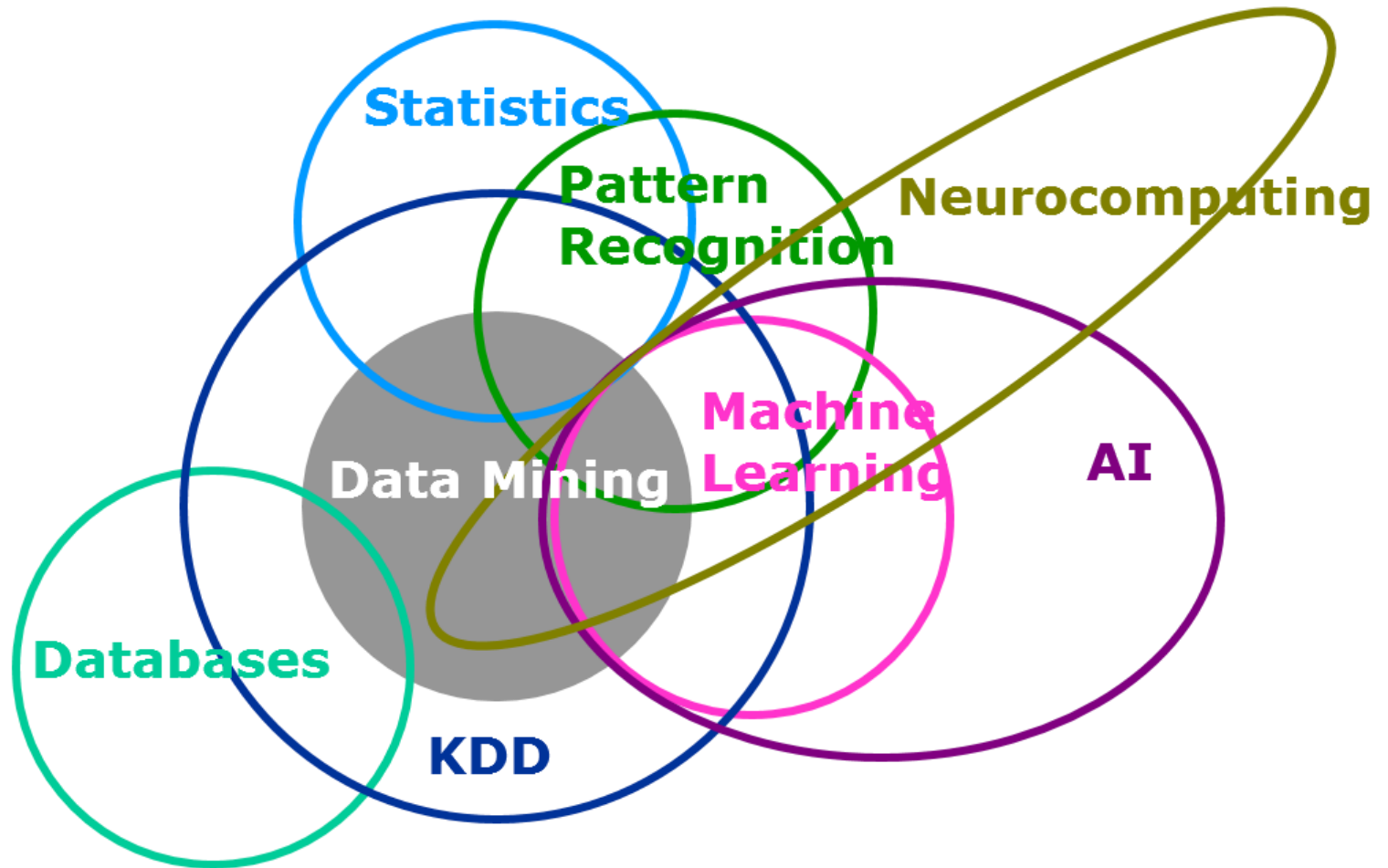
Data Mining and Foundation of AI (Good name) but you may feel a bit confusion.

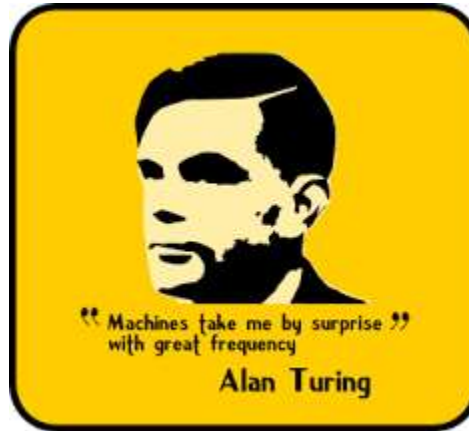
Actually, data mining partially belongs to AI field



How about MACHINE LEARNING then?

Relations between Data Mining, Machine Learning, AI and others





A bit of History of AI

A bit of history of AI (start)

Long long ago...

Philosopher Aristotle (385-322 B.C.)

My story will start from 1950 in the UK and one important publication:

Turing, A.M. (1950), Computing machinery and intelligence, Mind, Vol. 59, pp. 433-460.

M I N D
A QUARTERLY REVIEW
OF
PSYCHOLOGY AND PHILOSOPHY
— — — — —
I.—COMPUTING MACHINERY AND
INTELLIGENCE
By A. M. TURING



I propose to consider the question, 'Can machines think?' ...

A bit of history of AI (born)

The official birth of artificial intelligence:

- 1956: Dartmouth Conference - "Artificial Intelligence" adopted
- The term 'Artificial Intelligence' was coined in a proposal for the conference at Dartmouth College in 1956

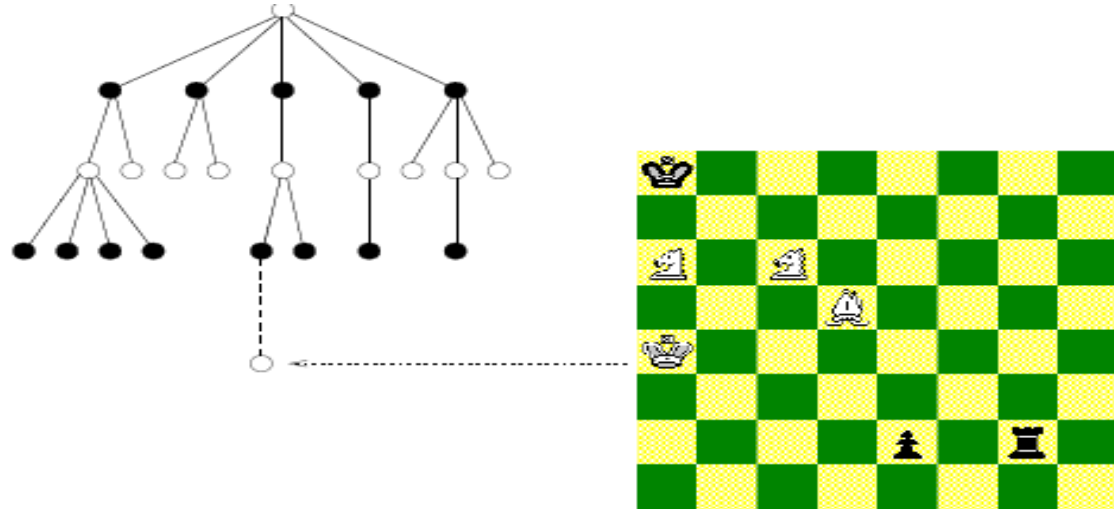
- 1) [Dr. Marvin Minsky](#) (MIT co-founder)
- 2) [Dr. Julian Bigelow](#) (Contri IT)
- 3) [Professor D.M. Mackay](#) (Neuro)
- 4) [Mr. Ray Solomonoff](#) (Probability)
- 5) [Mr. John Holland](#) (genetic)
- 6) [Mr. John McCarthy](#) (MIT, lisp)
- 7) [Dr. Claude Shannon](#) (IT)



A bit of history of AI (early days)

Two major themes that driven AI development from start to today are: Game-play and Robot

One of the early research in AI is search problem such as for game-playing. Game-playing can be usefully viewed as a search problem in a space defined by a fixed set of rules



Nodes are either white or black corresponding to reflect the adversaries' turns.

The tree of possible moves can be searched for favourable positions.

A bit of history of AI (victory)

- The real success of AI in game-playing was achieved much later after many years' effort.
- It has been shown that this search based approach works extremely well.
- In 1996 IBM Deep Blue beat Gary Kasparov for the first time. In 1997 an upgraded version won an entire match against the same opponent.

<https://www.kaggle.com/harryu/week10-reinforcement-learning-frozenlake-v0>



A bit of history of AI (development)

- **deductive logic (theorem-provers):**

All men are mortal

$\forall x (\text{man}(x) \rightarrow \text{mortal}(x))$

Socrates is a man

$\text{man}(\text{Socrates})$

Socrates is mortal

$\text{mortal}(\text{Socrates})$

- **Common sense reasoning:** the idea was to code common sense knowledge as logical axioms, and employ a theorem-prover (John McCarthy and Patrick Hayes)
- **Reasoning as search:** Newell and Simon developed a program called the "General Problem Solver".
- **Natural language Processing:** Ross Quillian proposed the semantic networks and Margaret Masterman & colleagues at Cambridge design semantic networks for machine translation

A bit of history of AI (golden years)

Golden years: 1956-1974

- **Lisp:** John McCarthy (MIT) invented the Lisp language.

Funding for AI research:

- Significant funding from both USA and UK governments

The optimism:

- 1965, Simon: "machines will be capable, within twenty years, of doing any work a man can do"
- 1970, Minsky: "In from three to eight years we will have a machine with the general intelligence of an average human being."

A bit of history of AI (1ST AI winter)

The first AI winter 1974–1980:

- **Problems**
 - **Limited computer power:** There was not enough memory or processing speed to accomplish anything truly useful
 - **Intractability and the combinatorial explosion.** In 1972 Richard Karp showed there are many problems that can probably only be solved in exponential time (in the size of the inputs), famously named as NP (nondeterministic polynomial time) complete/hard problem.
 - **Common sense knowledge and reasoning.** Many important applications like vision or natural language require simply enormous amounts of information about the world and handling uncertainty.
- **The end of funding**
 - The agencies which funded AI research became frustrated with the lack of progress and eventually cut off most funding for AI research.

A bit of history of AI (Rebranding)

Boom 1980–1987:

- In the 1980s a form of AI program called "expert systems" was adopted (rule-based technologies) by corporations around the world and knowledge representation became the focus of mainstream AI research
 - The power of expert systems came from the expert knowledge using **rules** that are derived from the domain experts
 - In 1980, an expert system called XCON was completed for the Digital Equipment Corporation. It was an enormous success: it was saving the company 40 million dollars annually by 1986
 - By 1985 the market for AI had reached over a billion dollars
- The money returns: the fifth generation project
- However, the fundamental issue about uncertainty (today we call it probability and fuzzy learning) and capability of dealing with big size data aren't really solved at all in that time.

A bit of history of AI (2nd AI winter)

the second AI winter 1987–1993

- In 1987, the Lisp Machine market was collapsed, as desktop computers from Apple and IBM had been steadily gaining speed and power and in 1987 they became more powerful than the more expensive Lisp machines made by Symbolics and others
- Eventually the earliest successful expert systems, such as XCON, proved too expensive to maintain, due to difficult to update and unable to learn.
- In the late 80s and early 90s, funding for AI has been deeply cut due to the limitations of the expert systems and the expectations for Japan's Fifth Generation Project not being met

A bit of history of AI (late 80s to 90s new ideas)

In the period of 2nd AI winter (in the late 80s), lots of new ideas are proposed which are the foundations of today's AI approaches.

- A completely new approach to AI, based on robotics, has been proposed by Brooks in his paper "Elephants Don't Play Chess", based on the belief that, to show real intelligence, a machine needs to have a body — it needs to perceive, move, survive and deal with the world.
- Artificial Neural Networks (ANN) to learn data features
- Probabilistic and Statistical approach
- Knowledge representation with Fuzzy framework
- Imaging processing and vision recognition

Winter again

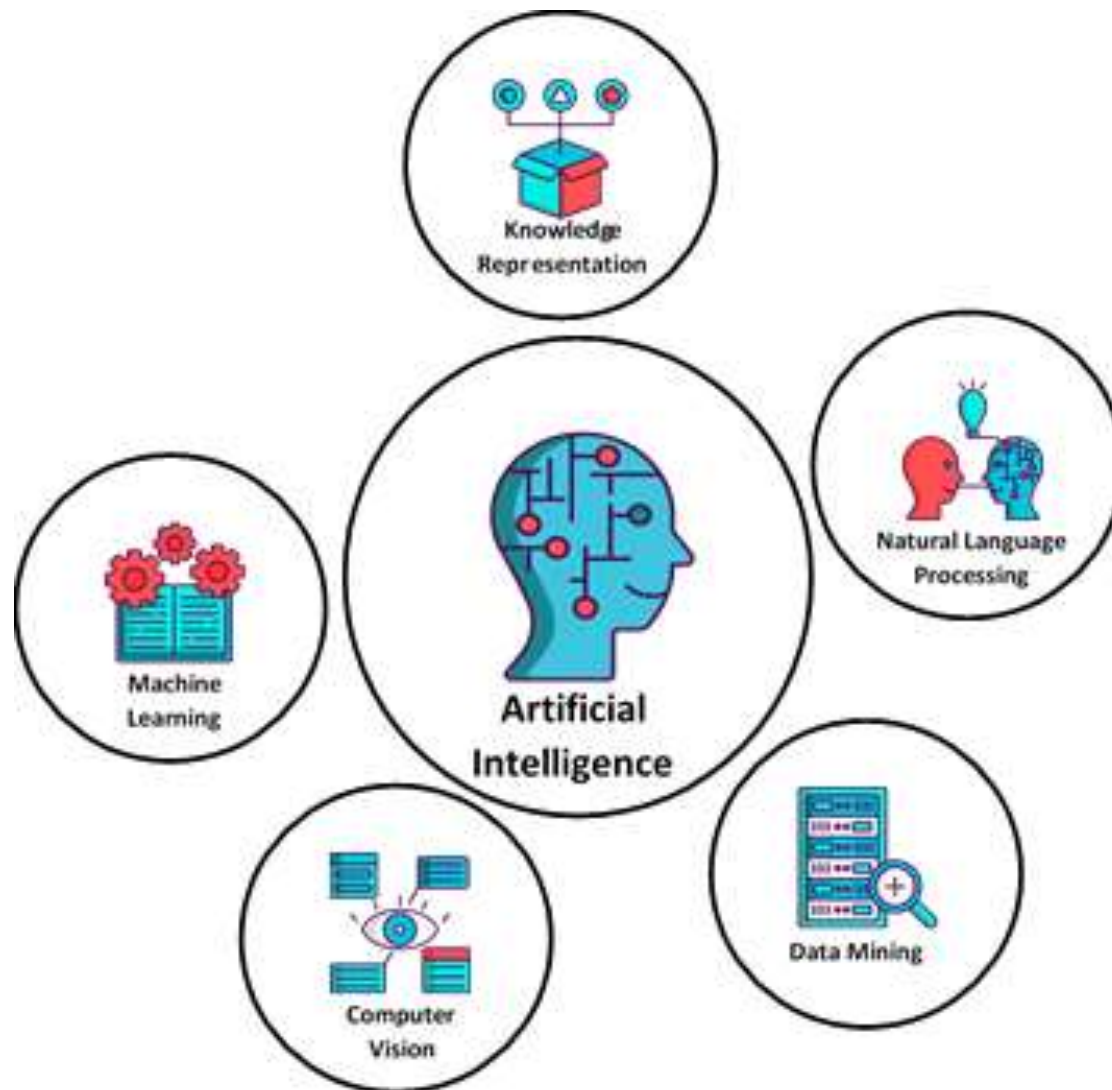


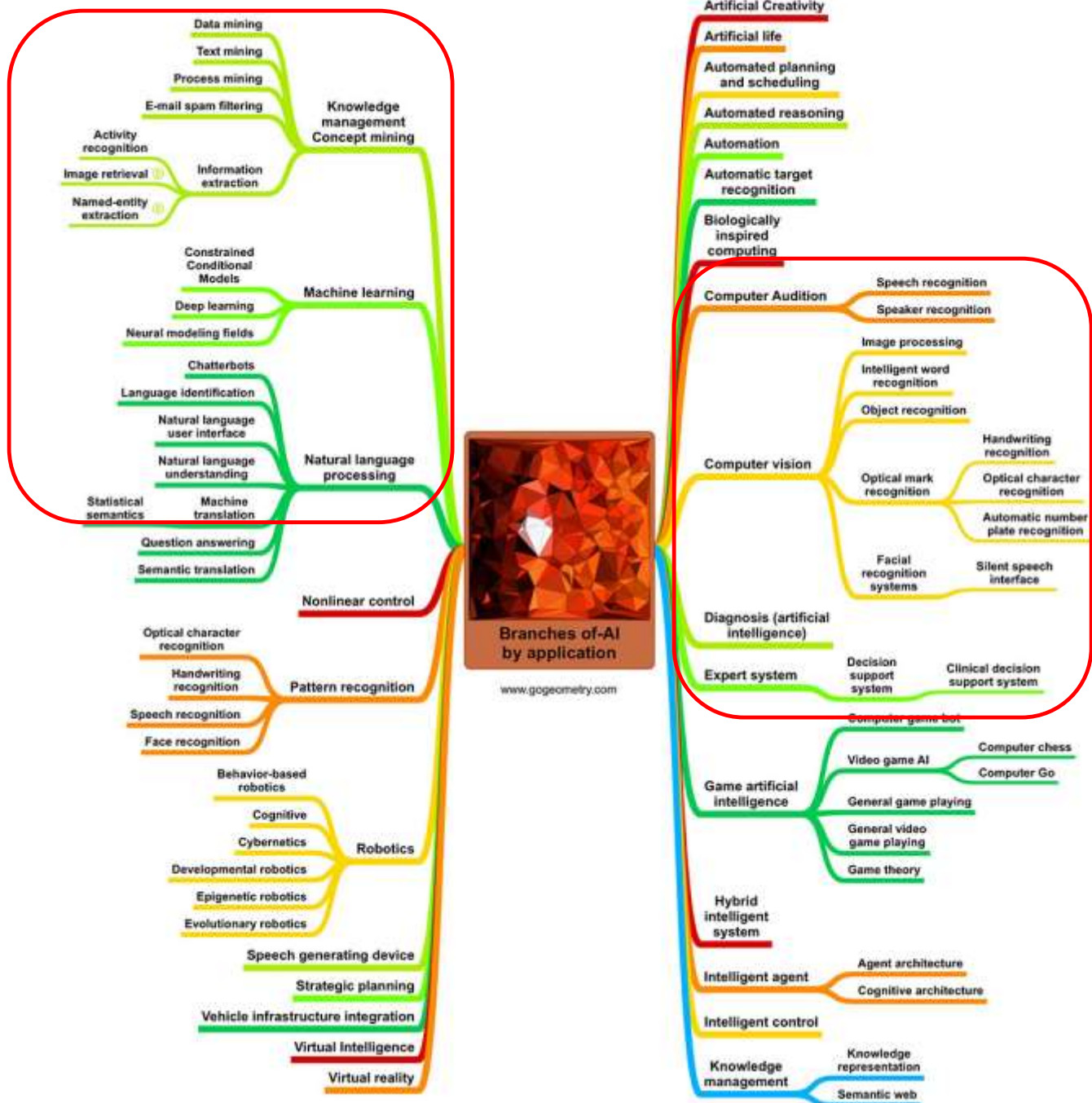
Brief history of AI (booming today)

Main reasons:

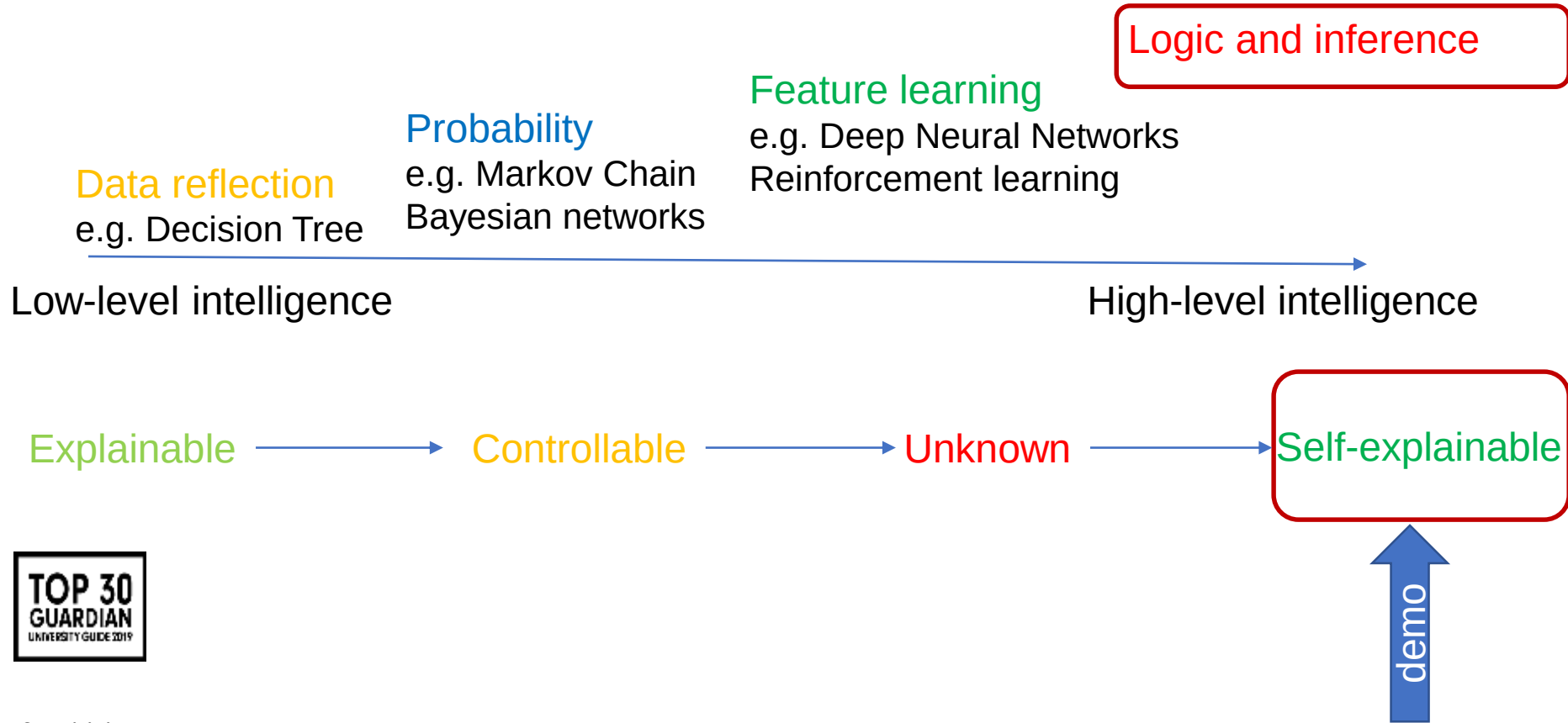
- Hardware power (GPU, storage)
- Cloud computing (Internet, IoT)
- Big Data (IoT, Mobil App, Social network, Linked data, activities on Internet)
- Industry driven (Google, Amazon, Facebook)
- the creation of new ties between AI and other fields working on similar problems
- A greater emphasis on solving specific sub-problems

Branches of AI technologies





Intelligent level



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→ A1 Deadline

→ A2 Deadline



Turing test

Turing test

In Alan Turing's 1950 paper, *Computing Machinery and Intelligence*, the mathematician posed the question: "Can machines think?" But almost immediately he dismissed that question as too "meaningless" to be worthy of discussion, and swapped it for the much-more specific: "Are there imaginable digital computers which would do well in the imitation game?"

Turing's "imitation game" was a simple party game with three players -- a man, a woman, and a judge of either sex. The judge sits in a room apart from the man and woman, and has to decide which of the pair is the person, and which is the machine.

<https://chatbotsmagazine.com/how-close-are-chatbots-to-pass-turing-test-33f27b18305e>

Demo <https://openai.com/blog/chatgpt/>

Demo <https://www.pandorabots.com/mitsuku/>

Summary

In this lecture, you takeaway following things:

1. Very basic concepts of Data Mining and AI.
2. The born of AI and its development history
3. Different areas of AI
4. The contents and assessment of the module

Next week

Lecture: Data processing and EDA

Practice: Python for deal with missing data and EDA diagrams

Practice, home work and further reading

1. Complete the first Python practice on Kaggle or Colab.
2. Try to understand the terms of Strong AI and Weak AI by reading these two articles (You need to answer a question in the discussion board):

<http://ceur-ws.org/Vol-2287/paper34.pdf>

<https://e3zine.com/strong-artificial-intelligence/>